第三章 训练大语言模型

- 知识回顾
- 预训练的重要性
- 大数据的重要性
- 人类价值观对齐
- 微调自己的LLM

1.4 相关知识回顾：模型训练（机器学习&梯度下降）

训练阶段（Training）

步骤1

一大堆数据

步骤3

选择最优模型

步骤2

一堆模型

$f_1, f_2, \dots, f_n$

- 输入：房价特征
- 输出：房价（标签）

- $y = \Sigma\alpha x + \beta$，$x$是特征值
- f由参数α和β定义

平均房间数
6.998
7.147

步骤4

最优模型f_{best}

房价（千美元）
33.4
36.2

测试阶段（Testing）

这也是所有机器学习方法的通用框架！

1.4 相关知识回顾：模型训练（机器学习&梯度下降）

回到找最优预测函数的问题 $w^* = arg\min_{w} L(w)$

Step 1: 计算一下 P_0 处的梯度 $\frac{d_L}{d_w}$

$$\frac{d_L}{dw} = \frac{d\left(\frac{1}{2N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2\right)}{dw}, \qquad \hat{y}_i = wx_i + b$$

$$= \frac{d\left(\frac{1}{2N}\sum_{i=1}^{N}\left((wx_i + b) - y_i\right)^2\right)}{d_w}$$

$$= \frac{1}{N}\sum_{i=1}^{N}(wx_i + b - y_i)x_i$$

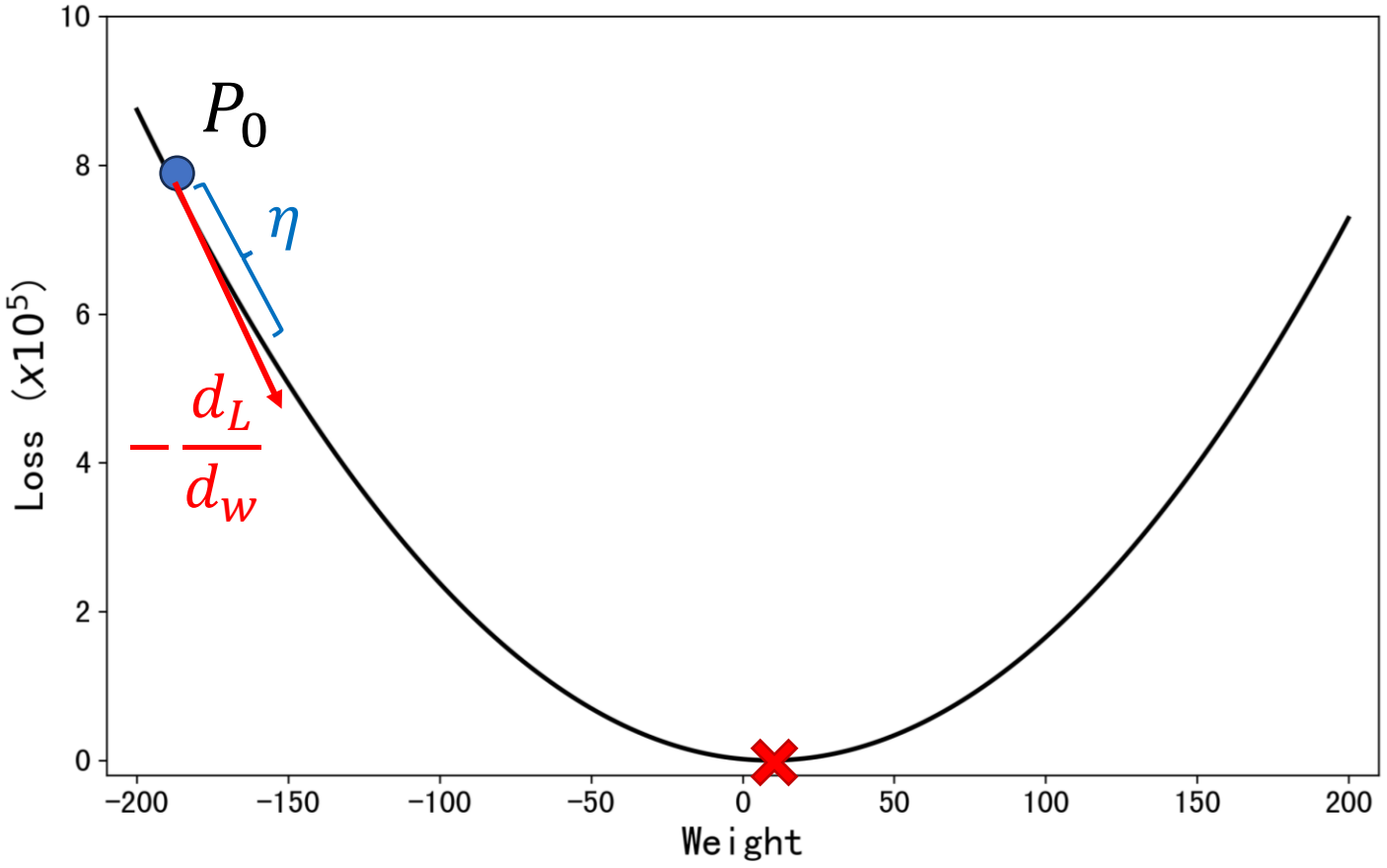
1.4 相关知识回顾：模型训练（机器学习&梯度下降）

回到找最优预测函数的问题 $w^* = arg\min_{w} L(w)$

Step 2: 沿梯度反方向$-\frac{d_L}{d_w}$移动 η

$$w = w - \frac{d_L}{d_w} \cdot \eta$$

学习率



1.4 相关知识回顾：模型训练（机器学习&梯度下降）

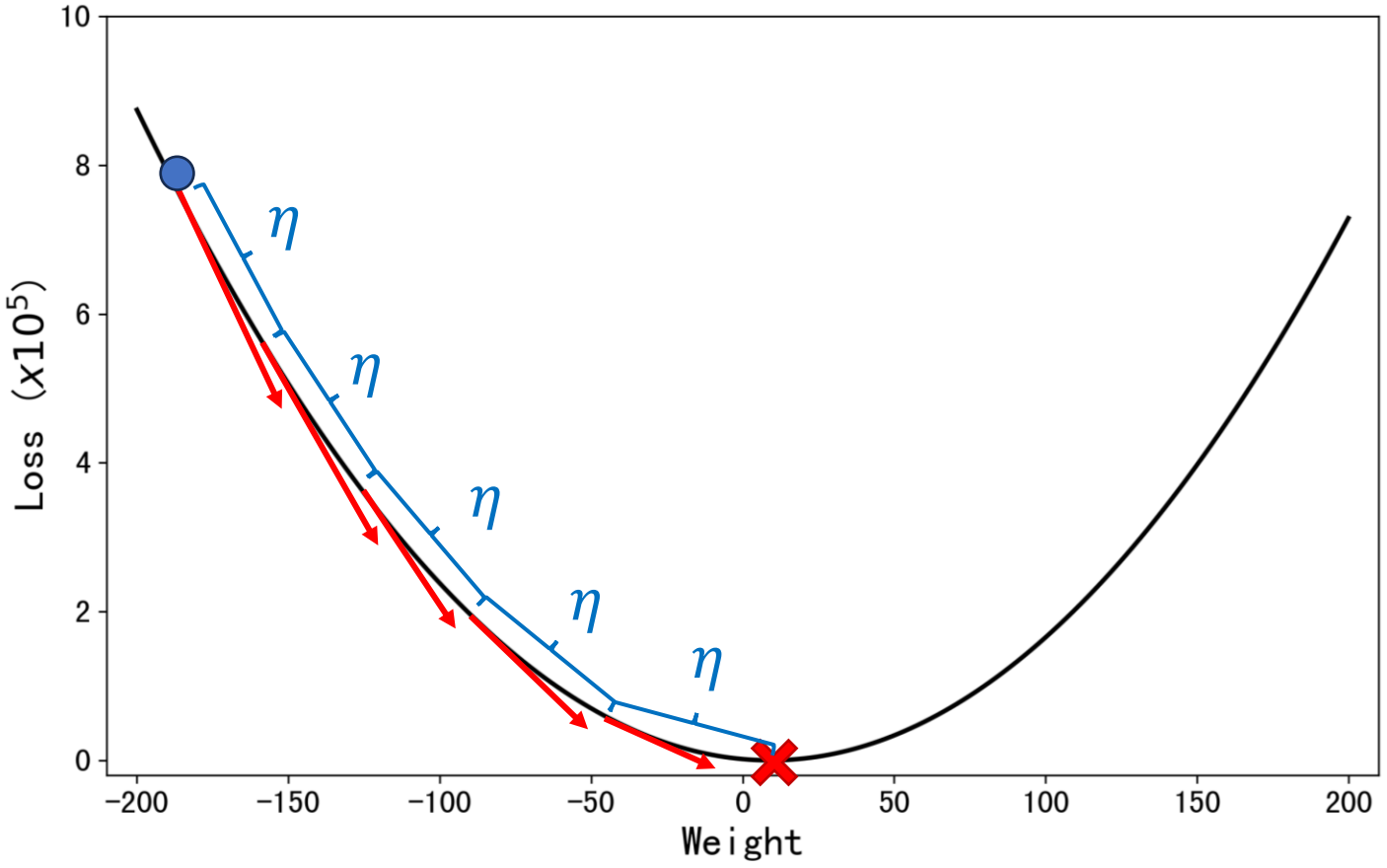
回到找最优预测函数的问题 $w^* = arg\min_{w} L(w)$

Step 3: 不断重复Step 1和Step 2

计算当前w的梯度 $\frac{d_L}{d_w}$

梯度下降

$$w = w - \frac{d_L}{d_w} \cdot \eta$$



1.4 相关知识回顾：模型训练（深度学习&反向传播）

步骤1

一大堆数据

步骤2

一堆模型

深度神经网络

步骤3

选择最优模型

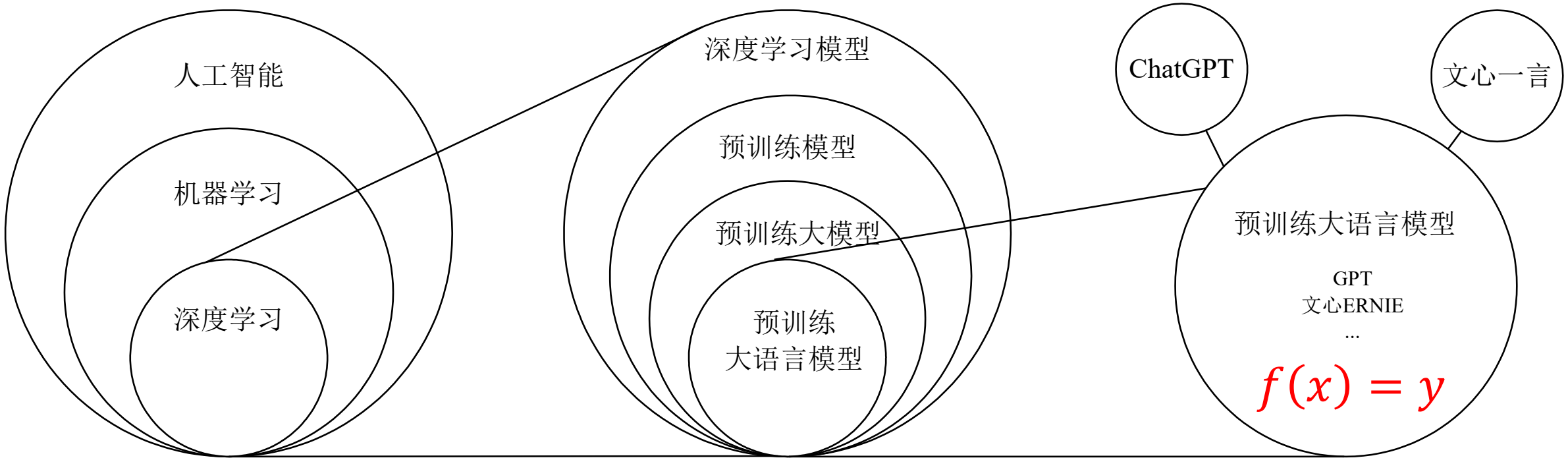
损失函数
梯度下降

《钟点工》2000年春节联欢晚会

I 打开冰箱　II 把大象塞进去　III 关上冰箱

1.5 相关知识回顾：大语言模型训练

- LLM与ML、DL使用类似的学习方法
- 问题在于“大”：量变引起质变

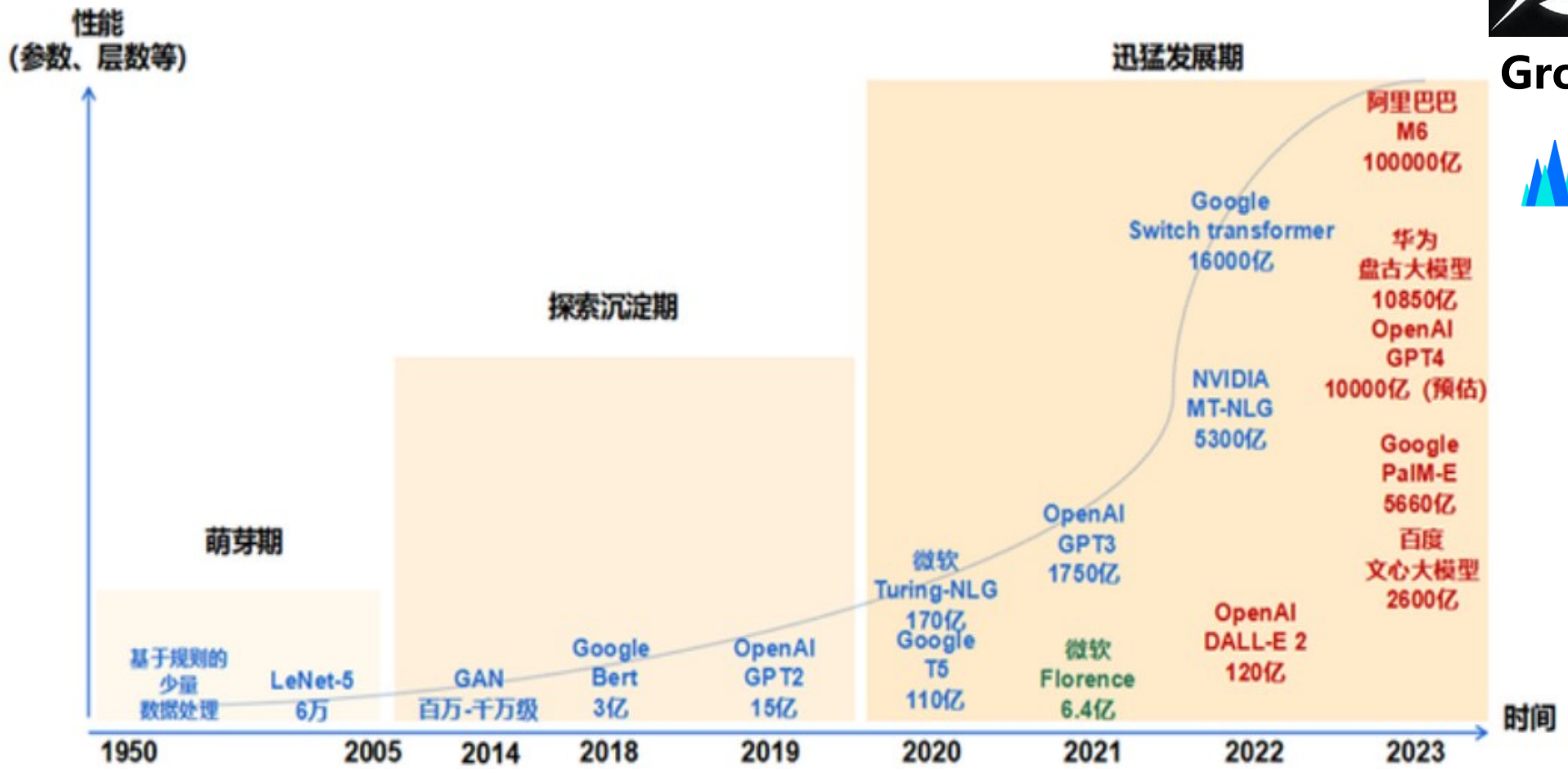


1.5 相关知识回顾：大语言模型训练

- 回顾ML课程中案例的数据特点

1.5 相关知识回顾：大语言模型训练

- 面对上万亿参数的通用模型，训练方式需要改变



Grok 4

GPT-5

Qwen3

火山引擎

Claude 3.7

1.5 相关知识回顾：大语言模型训练

模型体量：

GPT-1 (2018) 1.2亿

GPT-2 (2019) 15亿

GPT-3 (2020) 1750亿

（人工智能的天赋）

数据体量：

1GB

7000本书

40GB

580GB

（后天的努力）

1.5 相关知识回顾：大语言模型训练

Foundation Model
“没用”但是很重要

引入人类“教导”机器，称为**对齐（Alignment）**

第一阶段 自我学习，累计实力	第二阶段 名师指点，发挥潜力	第三阶段 参与实战，打磨技巧
Self-supervised Learning **自监督学习**	**Supervised Learning** **监督学习**	**RLHF** **基于人类反馈的强化学习**

“问题：鸣人最喜欢吃什么？答案：拉面。”

只问过程，不问结果

回答：好 | 坏

不问过程，只问结果

第三章 训练大语言模型

知识回顾

预训练的重要性

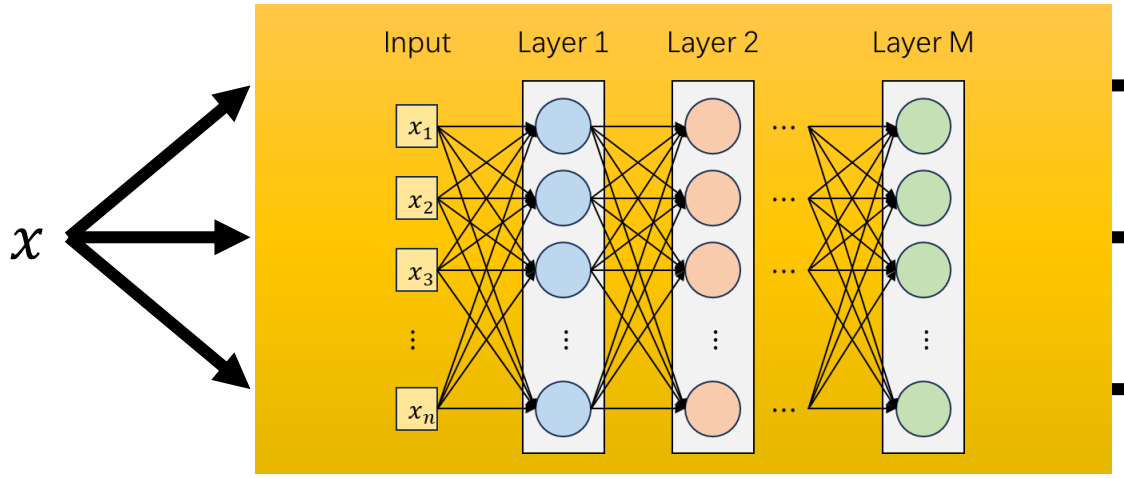
大数据的重要性

人类价值观对齐

微调自己的LLM

2.1 以前的模型如何训练?

Softmax回归 ➡ 深度神经网络



$\hat{y}$

$\hat{y}_1$

$\hat{y}_2$

$\hat{y}_3$

交叉熵
Cross Entropy Loss

y

y_1

y_2

y_3

人工进行数据标注

$$C = -\sum_{i=1}^{3} y_i ln\hat{y}_i$$

求偏导

$$\frac{\partial C}{\partial w_i}, \frac{\partial C}{\partial b_i}$$

反向传播
Backpropagation

更新参数

$\Delta w_1,$ Δb_1

$\Delta w_2,$ Δb_2

$\Delta w_3,$ Δb_3

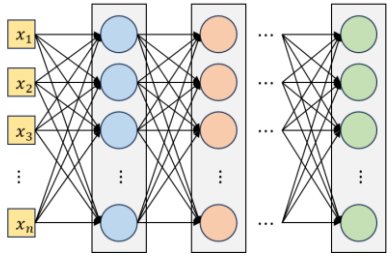
2.2 Supervised的方法能用于大模型吗?

- **以前的训练方法**

人工标注

X: 黄石最美的湖

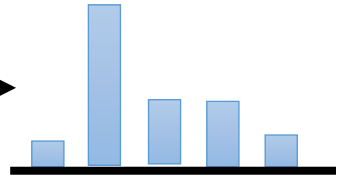
X: 黄石最美的大学



Y: 磁湖

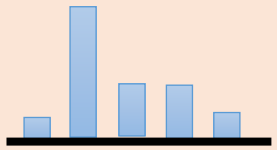
Y: 湖北理工学院

- **人工标注的问题**

使用者输入 → → 

"文字接龙"
一字一字的输出

"黄石最美的" → "湖"?
"黄石最美的" → "大"?

让模型自己"**边标注边学**" 

2.2 Supervised的方法能用于大模型吗?

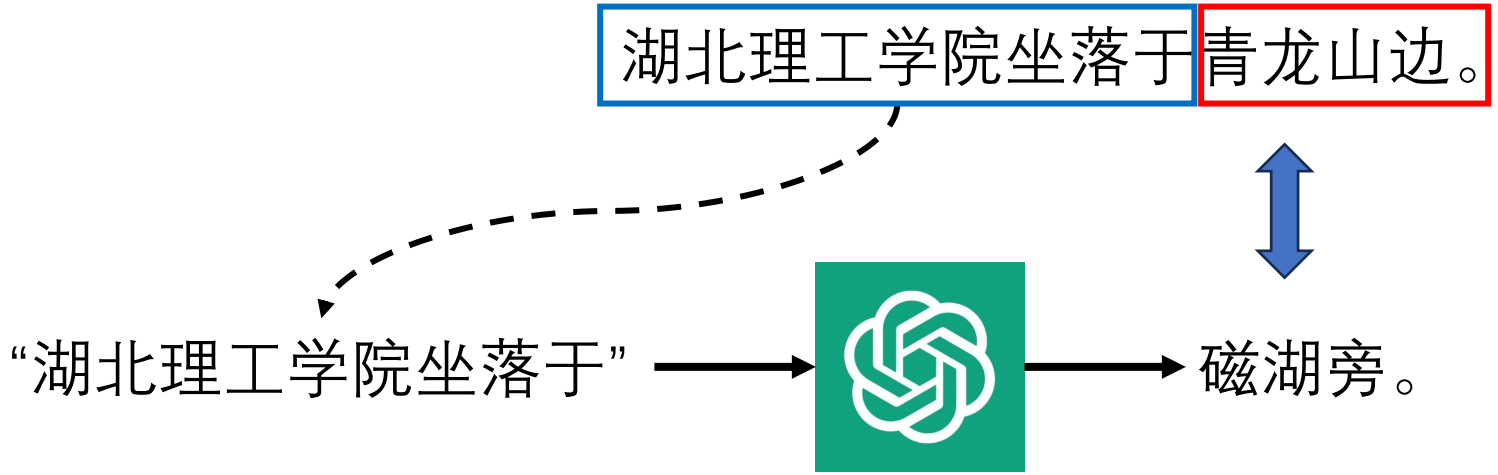
- 网络上每一段文字都可以教LLM做文字接龙

湖北理工学院坐落于青龙山边。

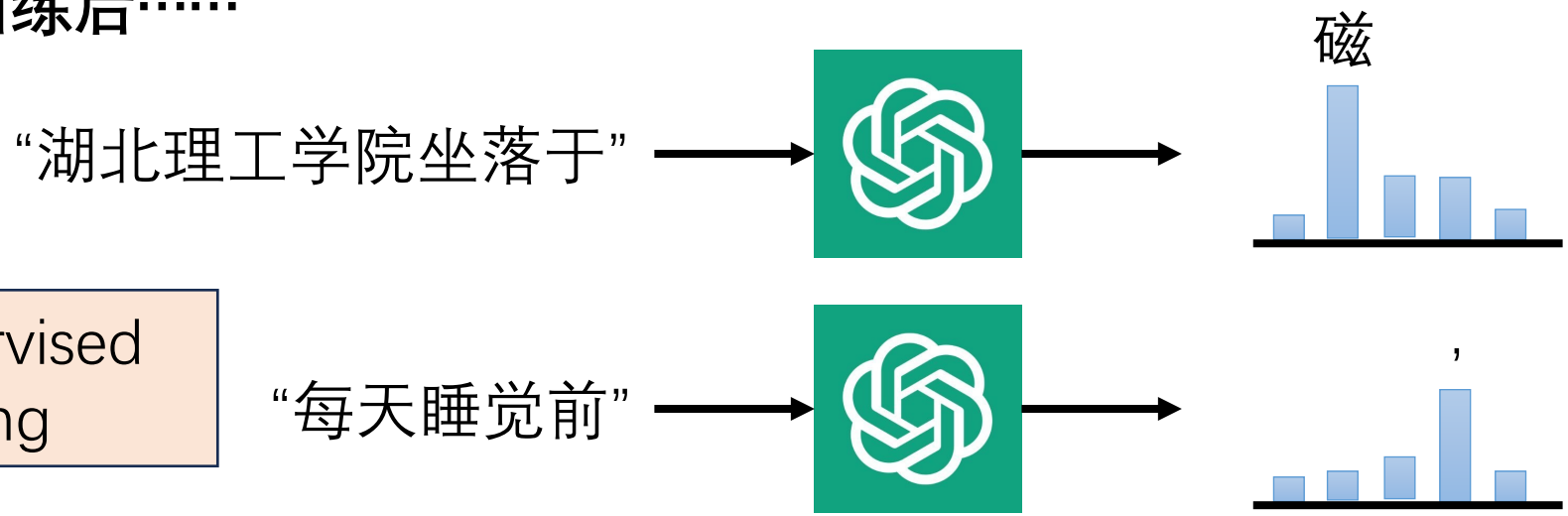
每天睡觉前，我一定要刷五分钟抖音。

2.2 Supervised的方法能用于大模型吗？

- **让模型自己学习**



- **经过大量训练后……**



Self-supervised Learning

2.2 Supervised的方法能用于大模型吗？

模型体量：

GPT-1 (2018)

1.2亿

GPT-2 (2019)

15亿

（人工智能的天赋）

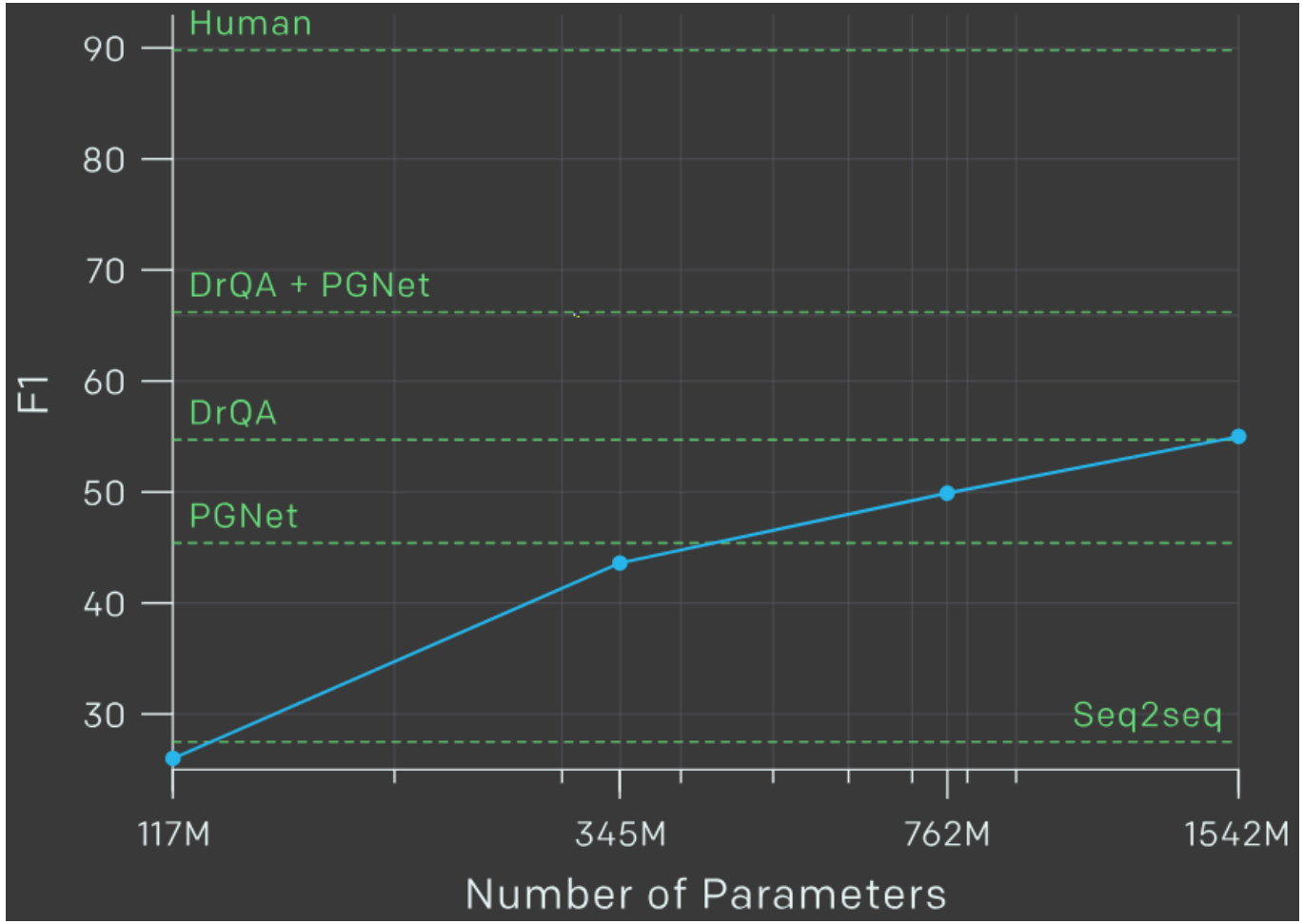
数据体量：

1GB

7000本书

40GB

（后天的努力）



https://openai.com/index/better-language-models/

2.3 预训练模型好像“用不了”

- **直接使用预训练的LLaMA-2-7b-base**

User：What is Machine Learning?

Assistant: Machine learning is a field of computer science that gives computers the ability to learn without being explicitly programmed. Machine learning is a subset of artificial intelligence.

Machine learning is a field of computer science that gives computers the ability to learn without being explicitly programmed. Machine learning is a subset of artificial intelligence.
…….

2.3 预训练模型好像“用不了”

- **使用微调后的LLaMA-2-7b-chat**

User：What is Machine Learning?

Assistant: Machine learning is a subfield of artificial intelligence (AI) that involves the use of algorithms and statistical models to enable machines to learn from data, make decisions, and improve their performance on a specific task over time.

Machine learning algorithms are designed to recognize patterns in data and learn from it, without being explicitly programmed to do so. The algorithms can be trained on large datasets, and as they process more data, they can make better predictions or decisions.
…….

2.3 预训练模型好像“用不了”

- 模型在微调之后的差异很大

Backbone Model	Template	Fine-tuned Modules	Fine-tuning Data	MT Bench
llama2-7b	one_shot	-	-	3.88
llama2-7b-chat	llama-2	-	-	6.27
llama2-7b	one_shot	lora	OpenWebText 80k	3.82
llama2-7b	one_shot	lora	llama2-7b-chat 80k	4.59
llama2-7b	one_shot	lora	gpt3.5-turbo 80k	4.93
llama2-7b-chat	llama-2	lora-base	gpt3.5-turbo 80k	6.60
llama2-7b-chat	llama-2	lora	gpt3.5-turbo 80k	6.33
llama2-7b	one_shot	lora	gpt4-turbo 80k	5.12
llama2-7b-chat	llama-2	lora-base	gpt4-turbo 80k	**6.71**
llama2-7b-chat	llama-2	lora	gpt4-turbo 80k	6.57
Mistral-7B-v0.1	zephyr	-	-	3.73
Mistral-7B-v0.1	zephyr	lora	undistilled 80k	3.57
Mistral-7B-v0.1	zephyr	lora	gpt4-turbo 80k	7.29
Mistral-7B-Instruct-v0.1	mistral	-	-	6.84
Mistral-7B-Instruct-v0.1	mistral	lora-base	gpt4-turbo 80k	6.54
Mistral-7B-Instruct-v0.1	mistral	lora	gpt4-turbo 80k	7.02
Mistral-7B-Instruct-v0.2	mistral	-	-	7.6
Mistral-7B-Instruct-v0.2	mistral	lora-base	gpt4-turbo 80k	7.46
Mistral-7B-Instruct-v0.2	mistral	lora	gpt4-turbo 80k	**7.74**
Meta-Llama-3-8b	llama-3	-	-	5.5
Meta-Llama-3-8b-Instruct	llama-3	-	-	7.86
Meta-Llama-3-8b	llama-3	lora	gpt4-turbo 80k	7.03
Meta-Llama-3-8b-Instruct	llama-3	lora	gpt4-turbo 80k	7.97
Meta-Llama-3-8b-Instruct	llama-3	lora-base	gpt4-turbo 80k	**8.21**
Meta-Llama-3-70b	llama-3	-	-	2.71
Meta-Llama-3-70b-Instruct	llama-3	-	-	8.63
Meta-Llama-3-70b	llama-3	lora	gpt4-turbo 80k	8.18
Meta-Llama-3-70b-Instruct	llama-3	lora	gpt4-turbo 80k	**9.03**
Meta-Llama-3-70b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.71

https://arxiv.org/abs/2409.00096

2.3 预训练模型好像“用不了”

- 模型在微调之后的差异很大

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llama2-7b	one_shot	lora	llama2-7b-chat 80k	4.59

Backbone Model	Template	Fine-tuned Modules	Fine-tuning Data	MT Bench
Meta-Llama-3-8b	llama-3	-	-	5.5
Meta-Llama-3-8b-Instruct	llama-3	-	-	7.86
Meta-Llama-3-8b	llama-3	lora	gpt4-turbo 80k	7.03
Meta-Llama-3-8b-Instruct	llama-3	lora	gpt4-turbo 80k	7.97
Meta-Llama-3-8b-Instruct	llama-3	lora-base	gpt4-turbo 80k	**8.21**
Meta-Llama-3-70b	llama-3	-	-	2.71
Meta-Llama-3-70b-Instruct	llama-3	-	-	8.63
Meta-Llama-3-70b	llama-3	lora	gpt4-turbo 80k	8.18
Meta-Llama-3-70b-Instruct	llama-3	lora	gpt4-turbo 80k	**9.03**
Meta-Llama-3-70b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.71

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https://arxiv.org/abs/2409.00096

2.4 对齐不需要大量的资料

Quality Is All You Need. Third-party SFT data is available from many different sources, but we found that many of these have insufficient diversity and quality — in particular for aligning LLMs towards dialogue-style instructions. As a result, we focused first on collecting several thousand examples of high-quality SFT data, as illustrated in Table 5. By setting aside millions of examples from third-party datasets and using fewer but higher-quality examples from our own vendor-based annotation efforts, our results notably improved. These findings are similar in spirit to Zhou et al. (2023), which also finds that a limited set of clean instruction-tuning data can be sufficient to reach a high level of quality. We found that SFT annotations in the order of tens of thousands was enough to achieve a high-quality result. We stopped annotating SFT after collecting a total of 27,540 annotations. Note that we do not include any Meta user data.

We also observed that different annotation platforms and vendors can result in markedly different downstream model performance, highlighting the importance of data checks even when using vendors to source annotations. To validate our data quality, we carefully examined a set of 180 examples, comparing the annotations provided by humans with the samples generated by the model through manual scrutiny. Surprisingly, we found that the outputs sampled from the resulting SFT model were often competitive with SFT data handwritten by human annotators, suggesting that we could reprioritize and devote more annotation effort to preference-based annotation for RLHF.

LLaMA 2 - https://arxiv.org/abs/2307.09288

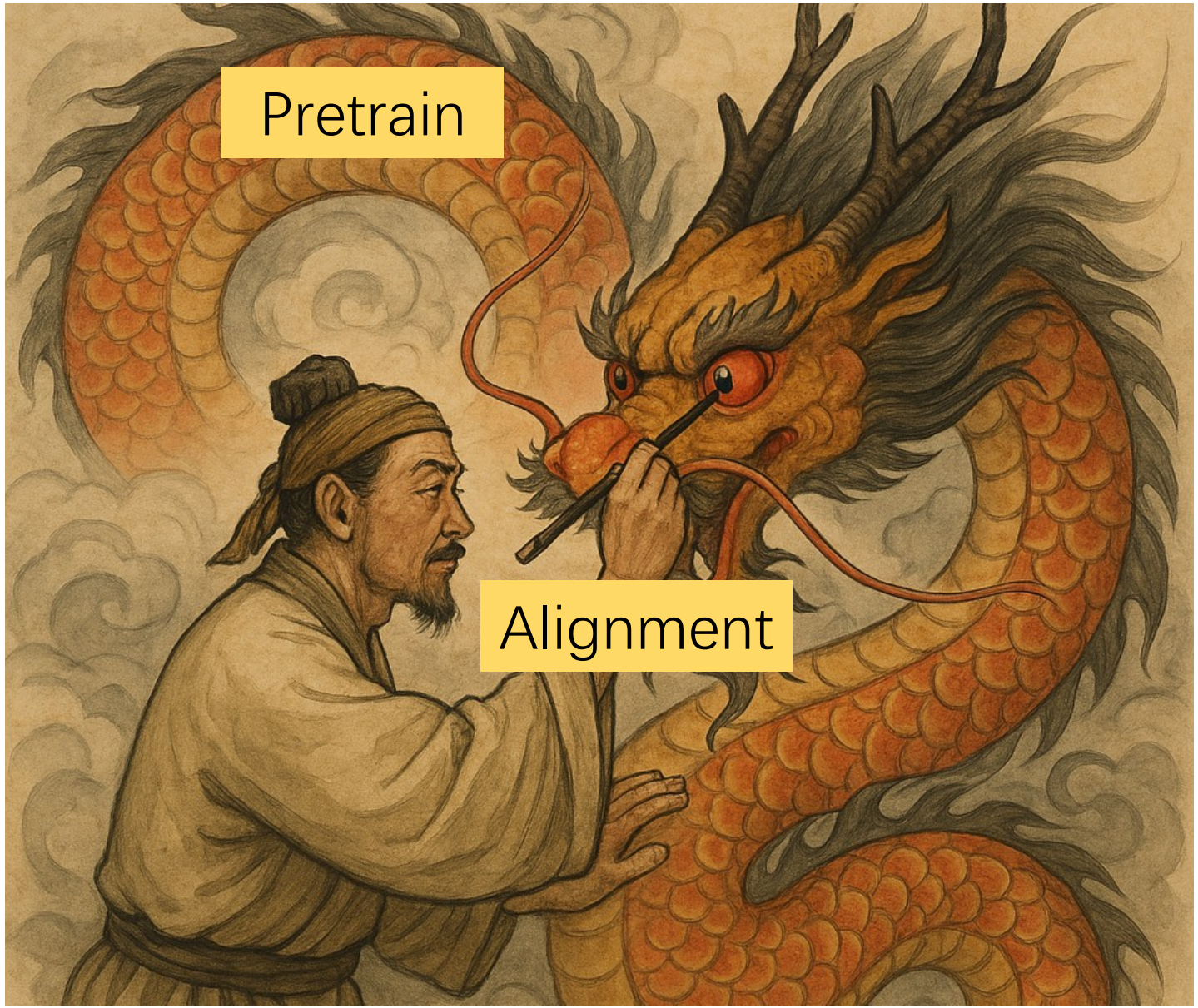
2.4 对齐不需要大量的资料

- **LIMA：Less Is More for Alignment**

Language models are pretrained to predict the next token at an incredible scale, allowing them to learn general-purpose representations that can be transferred to nearly any language understanding or generation task. To enable this transfer, various methods for *aligning* language models have thus been proposed, primarily focusing on *instruction tuning* [Mishra et al., 2021, Wei et al., 2022a, Sanh et al., 2022] over large multi-million-example datasets [Chung et al., 2022, Beeching et al., 2023, Köpf et al., 2023], and more recently *reinforcement learning from human feedback* (RLHF) [Bai et al., 2022a, Ouyang et al., 2022], collected over millions of interactions with human annotators. Existing alignment methods require significant amounts of compute and specialized data to achieve ChatGPT-level performance. However, we demonstrate that, given a strong pretrained language model, remarkably strong performance can be achieved by simply fine-tuning on 1,000 carefully curated training examples.

https://arxiv.org/abs/2305.11206

2.4 对齐不需要大量的资料，但是质量要高



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2.5 获得高质量的数据

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LLaMA 2 - https://arxiv.org/abs/2307.09288

2.5 获得高质量的数据

	Dataset	Open-QA	Brain.	CLS.	Gen.	Sum.	Rewrite	Closed-QA	Extract	Math	Code	Average
	Vanilla Models											
	Vanilla Qwen-2-7B	65.5	60.0	46.0	54.3	40.7	53.5	58.7	44.5	46.2	67.1	53.7
	Vanilla LLaMA-2-13B	1.4	3.8	5.0	1.0	6.7	17.5	12.2	13.6	0.0	17.1	6.9
	Qwen2-7B trained on different COIG-CQIA data source											
Qwen-2-7B	Zhihu	65.2	89.6	42.0	91.9	42.7	56.5	36.1	37.3	77.6	80.0	63.7
	Douban	53.8	67.3	15.0	68.1	13.3	34.0	37.8	27.3	81.0	43.6	47.0
	Xhs	49.3	60.0	12.5	42.9	13.3	12.0	31.7	16.4	71.4	27.1	36.9
	SegmentFault	53.8	68.5	41.5	69.0	33.3	74.5	48.7	42.7	76.2	65.7	58.6
	Ruozhiba	**77.6**	**95.8**	**64.5**	**96.7**	**76.7**	**91.5**	**82.6**	**72.3**	**90.5**	**87.1**	**83.5**
	Exam	51.4	83.8	54.2	75.2	30.7	73.0	72.2	57.3	49.5	71.4	62.9
	Logi QA	52.1	69.2	50.5	78.6	25.3	70.0	53.7	50.0	75.7	65.7	60.2
	WikiHow	48.3	28.5	1.0	41.9	20.7	5.0	20.9	12.7	62.4	47.9	30.2
	COIG PC	53.1	95.4	53.0	85.2	47.3	56.5	50.4	60.0	61.9	42.9	62.1
	Chinese Tra	41.7	73.1	41.0	79.5	28.7	69.5	55.2	41.8	80.0	58.6	58.2
	Human Value	65.5	90.0	60.5	86.7	58.0	85.0	64.8	50.9	78.6	72.9	72.8
	COIG-CQIA-Fullset	63.8	88.3	55.0	92.9	51.0	59.0	67.8	64.5	66.7	65.7	68.7
	COIG-CQIA-Subset	59.7	86.2	54.0	91.9	54.3	58.5	68.3	70.9	83.3	71.4	70.3

https://arxiv.org/abs/2403.18058

2.5 获得高质量的数据

- 弱智吧

ModelScope 首页 模型库 数据集 创空间 AIGC专区 文档 社区 MCP广场 GitHub

弱智吧 AI-ModelScope / ruoz…

数据集介绍 数据预览 数据集文件 快速使用 交流反馈 2

Cherrytest add 5305235


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{"author": "公孙闲", "content": "343.法国工人开始思考，如果一切都停业了，那我们怎么举行罢工。", "l_num": 49, "ctime": "2021-05-09 02:31"},
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{"author": "公孙闲", "content": "341.会不会我已经死了，而大家都瞒着我呢", "l_num": 301, "ctime": "2021-05-09 12:39"},
{"author": "公孙闲", "content": "340.面对案发现场留下的指纹，有洁癖的王警官将其一一擦干净了。", "l_num": 302, "ctime": "2021-05-09 12:40"},
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{"author": "公孙闲", "content": "291.熬夜只是今天没过够，向明天要时间。", "l_num": 27, "ctime": "2022-01-27 14:34"},
{"author": "公孙闲", "content": "290.“我是free控。”艾伦说道", "l_num": 28, "ctime": "2022-01-27 14:34"},
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{"author": "公孙闲", "content": "289.如果有人批评你不够客观，那么很可能是你的观点不符合他的主观", "l_num": 115, "ctime": "2022-01-27 17:23"},
{"author": "公孙闲", "content": "288.心脏明明是最强壮的肌肉，为什么大家的心都容易受伤呢", "l_num": 117, "ctime": "2022-01-27 17:23"},
```

2.5 获得高质量的数据

- 弱智吧

这台手术很成功，我失败的人生结束了。

楼主

张脑丸

怀念过去是不是在时间的长河里刻舟求剑

楼主

尔多龙听不见

一个人如果没有梦想...

楼主

春华

跟无忧无虑有什么区别呢?

2.5 获得高质量的数据

acteristics contributed to its effectiveness. The Ruozhiba dataset has inherent logical structures, includes cognitive and linguistic traps, and features jokes and riddles, as well as artistic and abstract rhetorical techniques. These elements, in turn, challenge the model' s multi-hop reasoning capabilities, enhancing its understanding of the Chinese language during fine-tuning and improving its capacity for complex logical reasoning. Human Value ranks second on average across all subsets, which aligns with expectations, as this subset contains a substantial amount of high-quality human-annotated data that aligns well with human values. This data not only improved instruction-following capabilities during fine-tuning but also prevented models from biasing towards specific values, enhancing universality. Moreover, WikiHow scores only 30.2 on

Ruozhiba is a sub-forum within Baidu Tieba, China's largest interest-based online community platform. This particular forum is renowned for its linguistic complexity, featuring posts rich in wordplay, including puns, polysemous terms, causal reversals, and homophones. Many of these posts are ingeniously crafted with logical traps that present cognitive challenges even for native speakers. In our study, we focus on the 500 most upvoted threads in this forum. We used the thread titles as potential instructions, carefully filtering out those that were non-instructive (such as mere declarative statements or unanswerable queries) or contained toxic content. For answer curation, human evaluators first identified the traps within the instructions and then prompted GPT-4 to generate responses. This process was repeated until GPT-4 produced correct answers.

https://arxiv.org/abs/2403.18058

2.5 获得高质量的数据

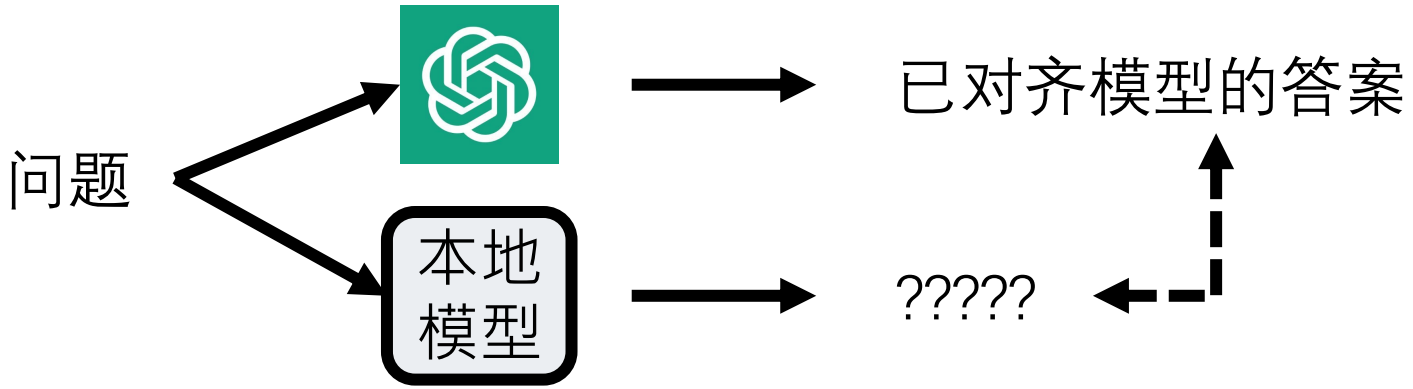
Source	Type	Description	Quantity	Data Processing
Zhihu (A.1)	Forum	Comprehensive Q&A platform	8837	🟦🟩🟥🟪
Segment Fault (A.1)	Forum	Tech community for IT developers	458	🟦🟩🟪
Douban (A.1)	Social Media	User-driven platform focused on literature and the arts	3132	🟦🟩🟧🟪
Xiaohongshu (A.1)	Social Media	Life experiences sharing platform	1508	🟦🟩🟧🟪
Ruozhiba (A.1)	Forum	Tieba[1] subcommunity interested in logical traps.	240	🟦🟩🟥🟪
Encyclopedia Article	World Knowledge	Comprehensive encyclopedic knowledge from various website	20020	🟦🟩🟧🟪
Encyclopedia of China(A.2.1)	World Knowledge	Comprehensive Chinese encyclopedia	1706	🟦🟩🟧🟪
WikiHow (A.2.1)	World Knowledge	Step-by-step guides and how-tos	1876	🟦🟩🟪
Medical Ariticle	World Knowledge	Health-related knowledge	186	🟦🟩🟪
Middle&High School Exam(A.3)	Exam	Standardized exam for middle&high school students	2000	🟦🟩🟪
Graduate Entrance Exam(A.3)	Exam	National graduate entrance exam	475	🟦🟩🟪
Logical Exam(A.3)	Exam	Logistic reasoning exam questions	422	🟦🟩🟪
Law Exam	Exam	Law graduate entrance exam questions	2645	🟦🟩🟪
COIG PC(A.4)	NLP Dataset	A massive dataset for instruction fine-tuning	3000	🟩🟪
COIG-Human-Value(A.4)	NLP Dataset	Value-related tasks from COIG	101	🟩🟪
CValues(A.4)	NLP Dataset	Detoxifying answers written by experts for harmful questions	906	🟪
Chinese Traditional (A.3)	NLP Dataset	Chinese traditional culture tasks from various datasets	1111	🟦🟩🟪
Finance NLP Task	NLP Dataset	NLP tasks in the financial domain	600	🟩🟪
Total			**45173**	

Table 1: Overview of different data sources. We list each data source's source type, description, quantity, and data processing stages. The colored squares represent different stages we applied to the data processing: 🟦 Raw Data Collection; 🟩 Rule-based filter; 🟧 Template Curation; 🟥 Model-Based Filter; 🟪 Human Finalization.

https://arxiv.org/abs/2403.18058

2.5 获得高质量的数据

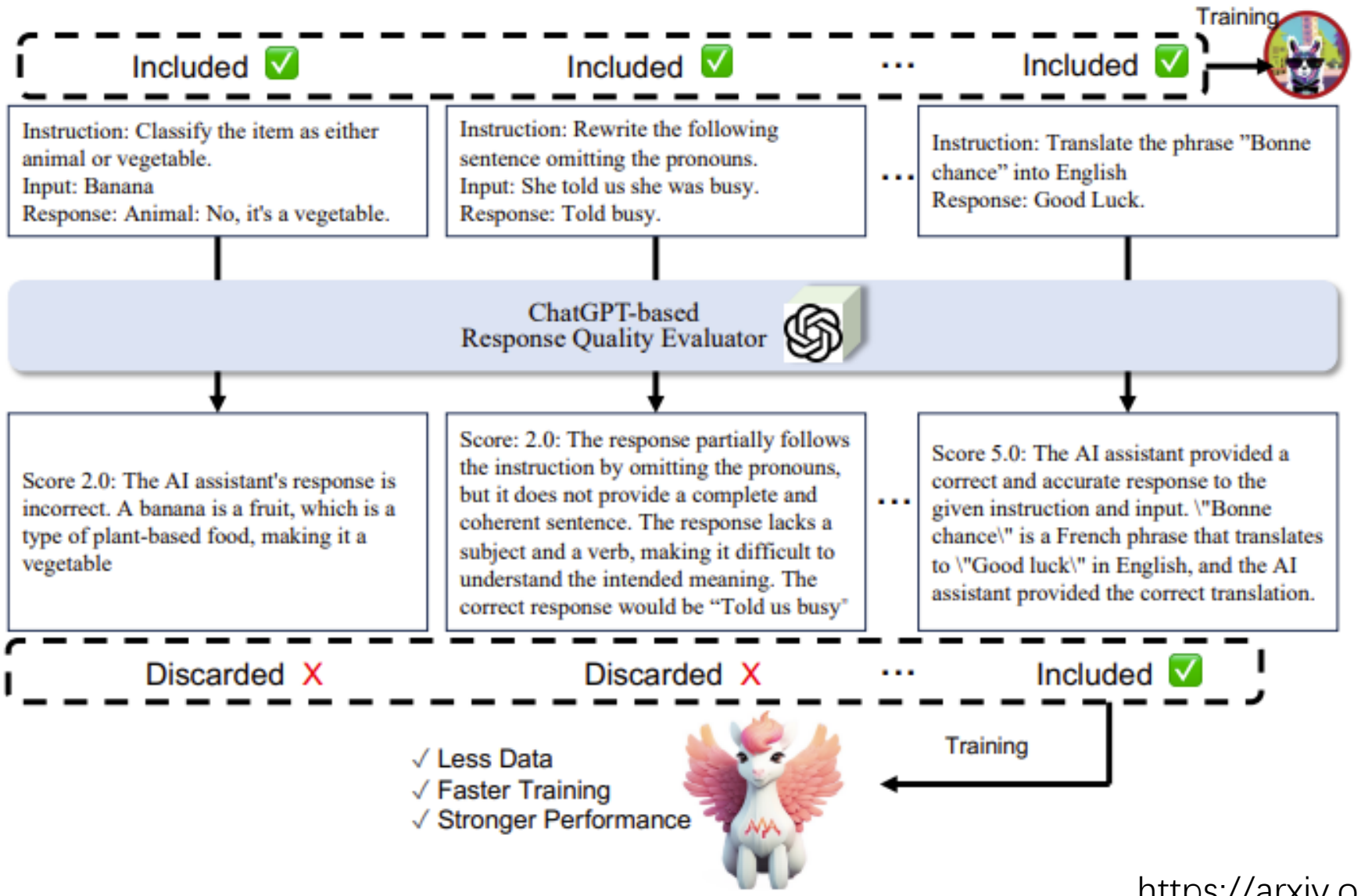
- 知识蒸馏（Knowledge Distillation）



	Student	Teacher	Data	Cost
Alpaca	LLaMA1-7B-base	ChatGPT	52k	$100
Vicuna	LLaMA1-7B-base	ChatGPT	70k	$140
Sky-T1	Qwen2.5-32B-Instruct	QwQ	17k	$450
S1	Qwen2.5-32B-Instruct	Gemini	1k	<$50

https://arxiv.org/abs/2307.08701

2.5 获得高质量的数据

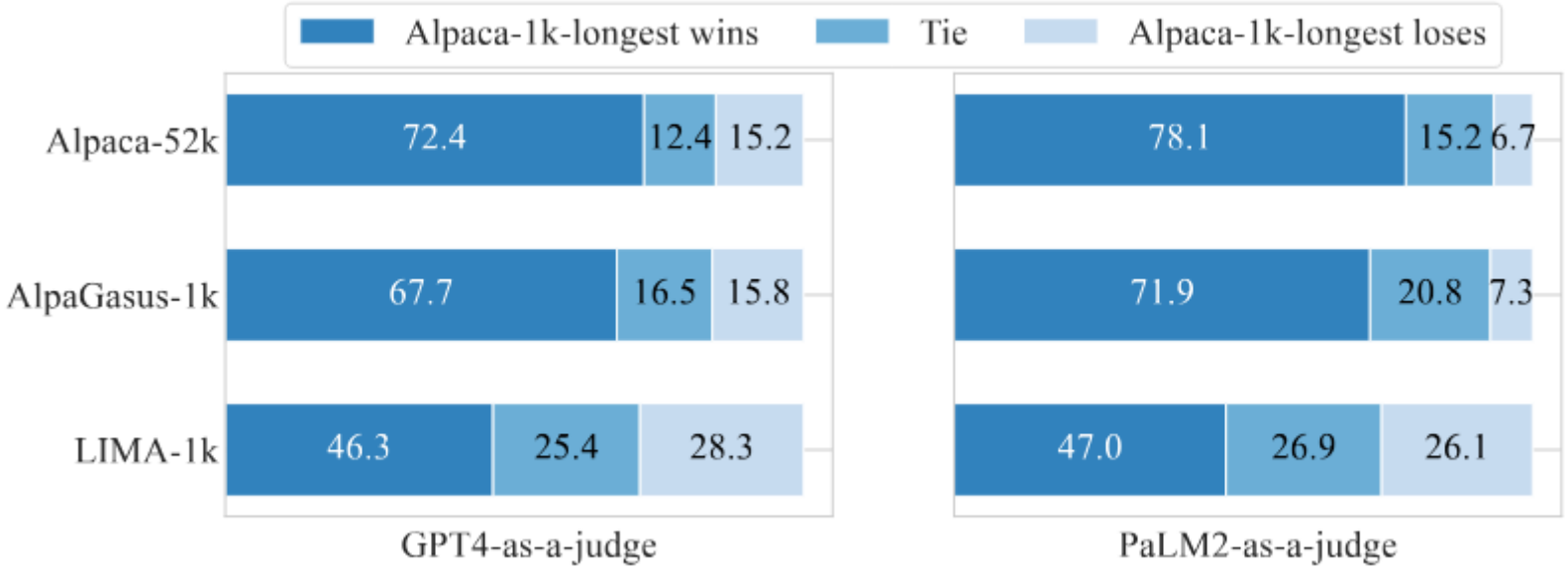


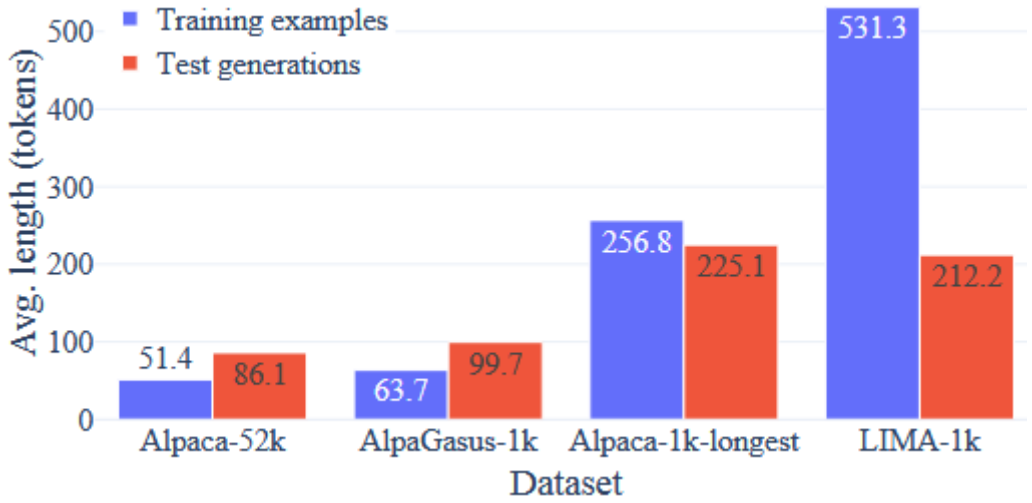
https://arxiv.org/abs/2307.08701

2.5 获得高质量的数据

- **试试选最长的资料**

是不是裁判模型就喜欢长的答案？



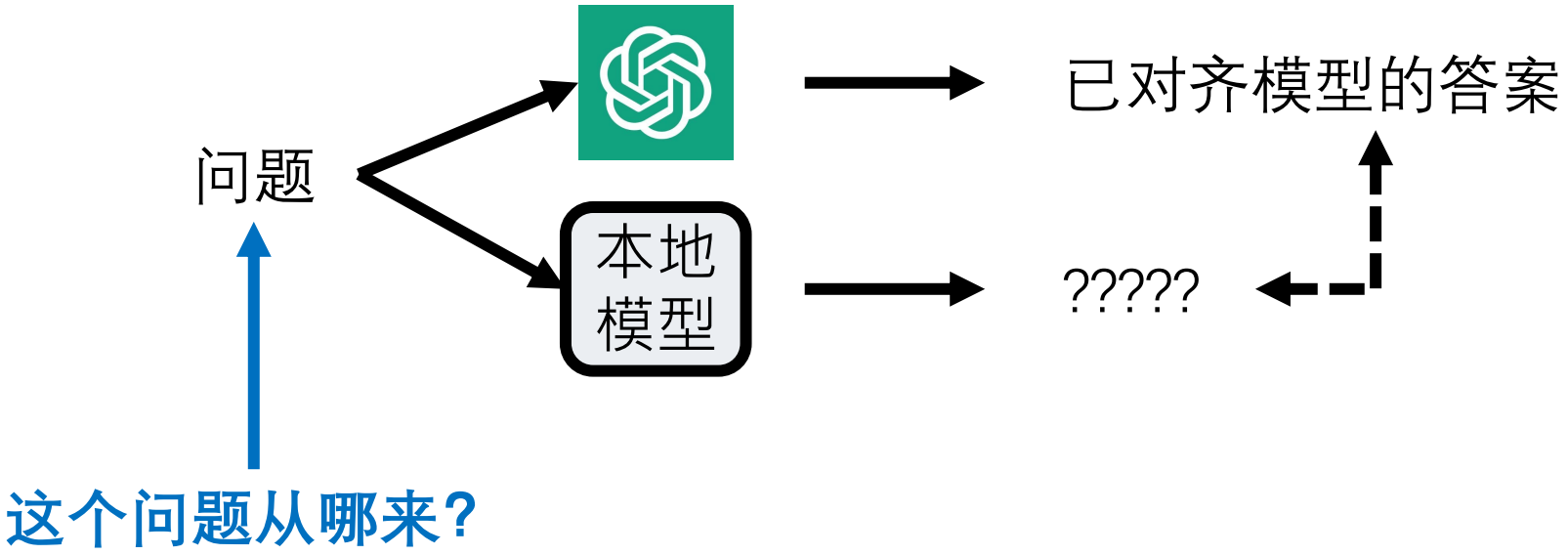


LIMA用的资料最长，但是提升反而最差

https://arxiv.org/abs/2402.04833

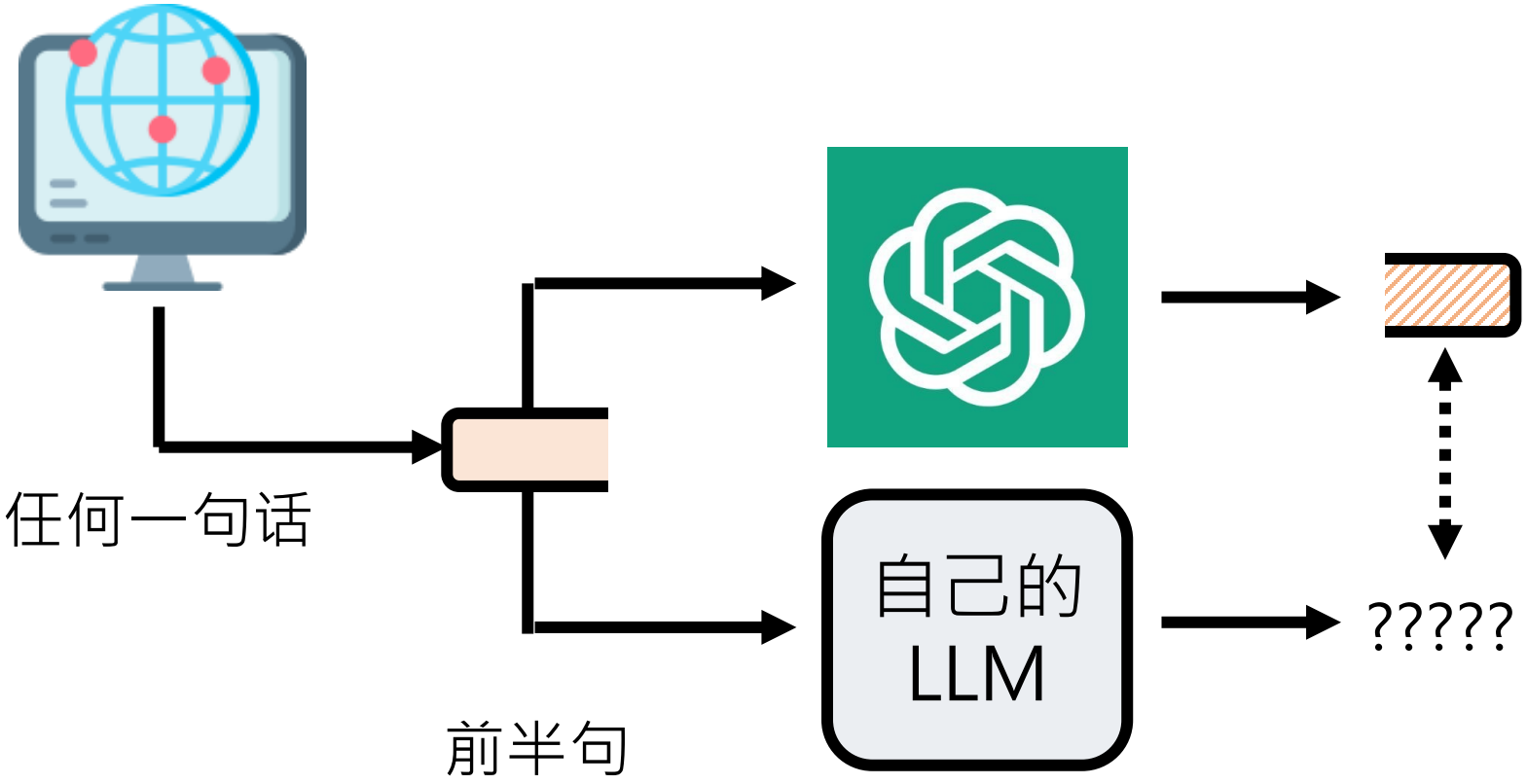
2.5 获得高质量的数据

- 知识蒸馏（Knowledge Distillation）



2.5 获得高质量的数据

- 知识蒸馏（Knowledge Distillation）



https://arxiv.org/abs/2409.00096

2.5 获得高质量的数据

- **知识蒸馏（Knowledge Distillation）**

✓ 上半句: …… The nondiscrimination policy seeks to ensure employers with more than 10 employees

✓ 原下半句: in the city as well as those who provide housing and public accommodations ……

✓ ChatGPT续写: , as well as housing providers, public accommodations, and city contractors, do not discriminate based on ……

✓ 上半句: …… Davis was recently hired as a morning anchor for CBS46. She is scheduled to

✓ 原下半句: start Jan. 2. ……

✓ ChatGPT续写: begin her new role despite the recent arrest. ……

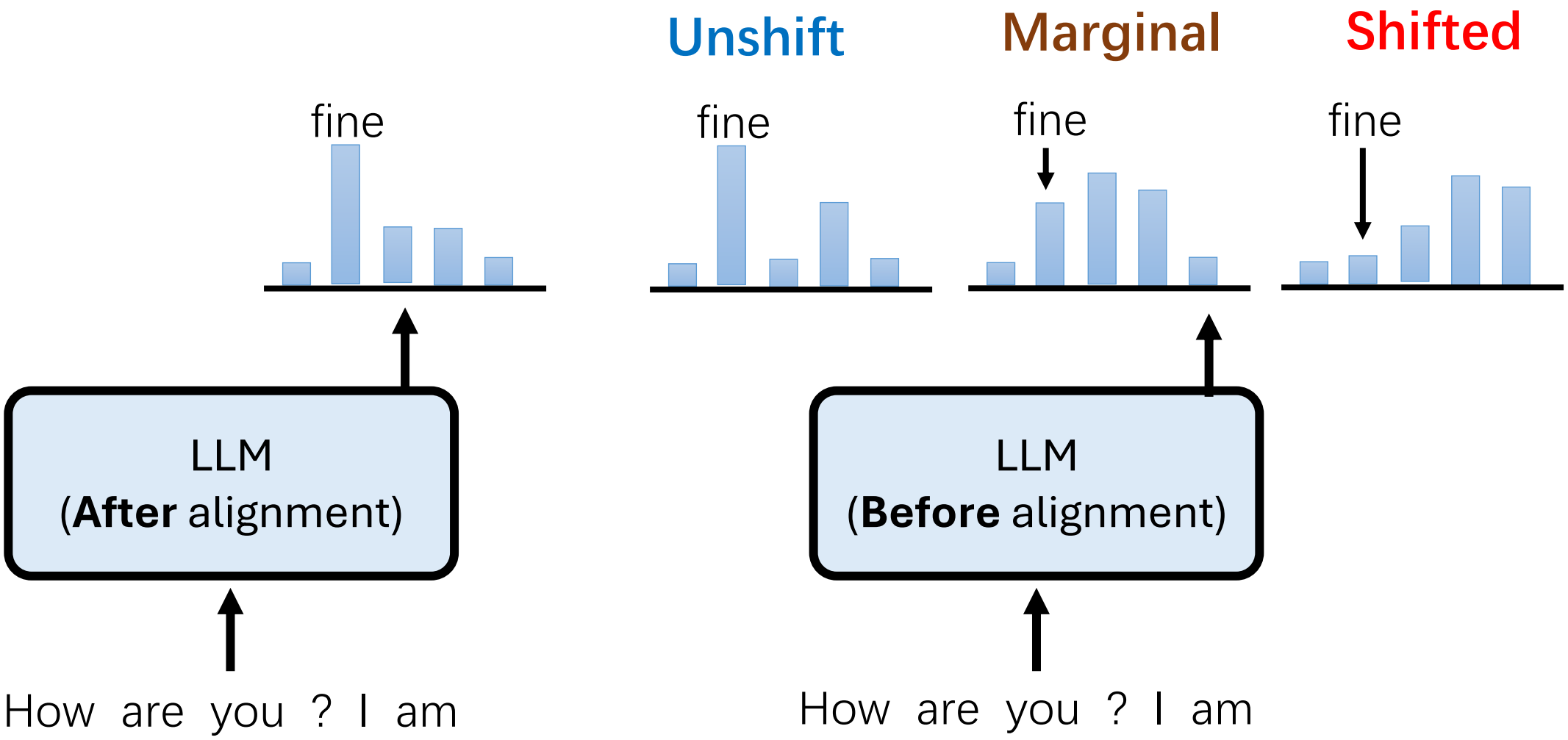
https://arxiv.org/abs/2409.00096

2.5 获得高质量的数据

Backbone Model	Template	Fine-tuned Modules	Fine-tuning Data	MT Bench
Meta-Llama-3-8b	llama-3	-	-	5.5
Meta-Llama-3-8b-Instruct	llama-3	-	-	7.86
Meta-Llama-3-8b	llama-3	lora	gpt4-turbo 80k	7.03
Meta-Llama-3-8b-Instruct	llama-3	lora	gpt4-turbo 80k	7.97
Meta-Llama-3-8b-Instruct	llama-3	lora-base	gpt4-turbo 80k	**8.21**
Meta-Llama-3-70b	llama-3	-	-	2.71
Meta-Llama-3-70b-Instruct	llama-3	-	-	8.63
Meta-Llama-3-70b	llama-3	lora	gpt4-turbo 80k	8.18
Meta-Llama-3-70b-Instruct	llama-3	lora	gpt4-turbo 80k	**9.03**
Meta-Llama-3-70b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.71

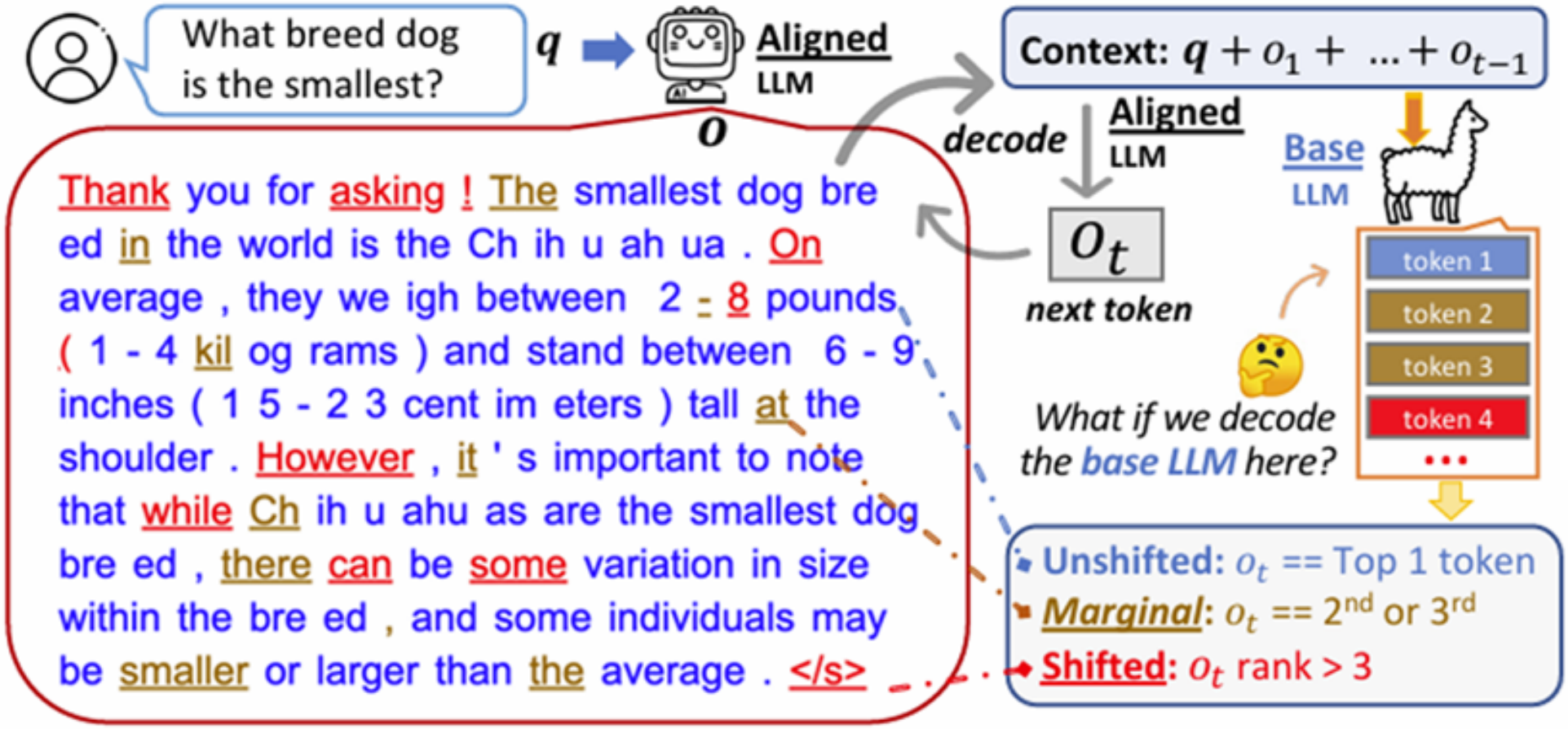
https://arxiv.org/abs/2409.00096

2.6 其实对齐的作用并没有那么大



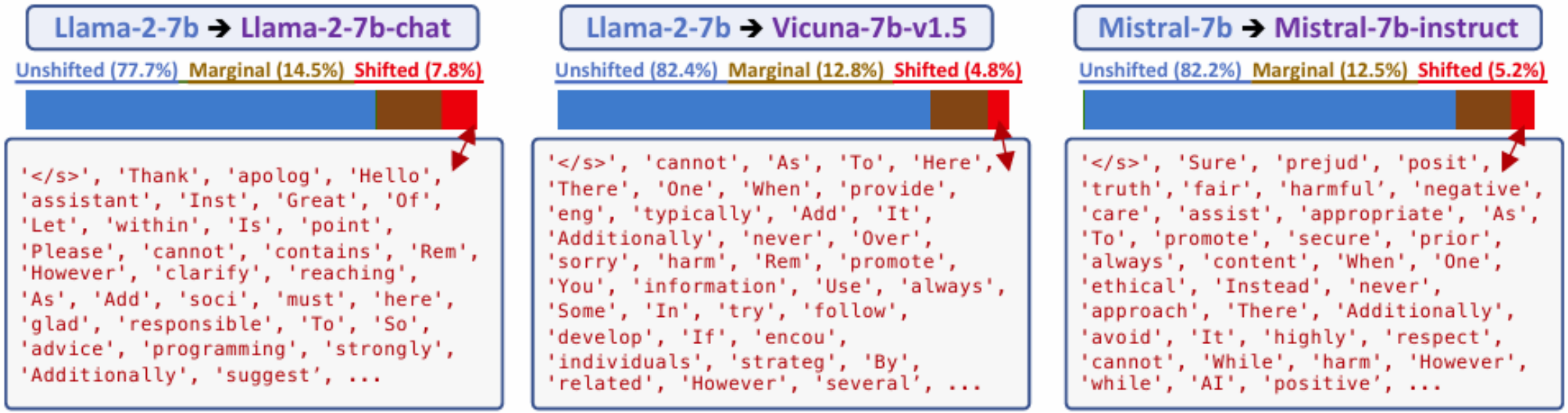
https://arxiv.org/abs/2312.01552

2.6 其实对齐的作用并没有那么大



https://arxiv.org/abs/2312.01552

2.6 其实对齐的作用并没有那么大



仅仅是一些符号、连接词、语气词改变了，对内容本身好像没有任何提升？

https://arxiv.org/abs/2312.01552

2.6 其实对齐的作用并没有那么大

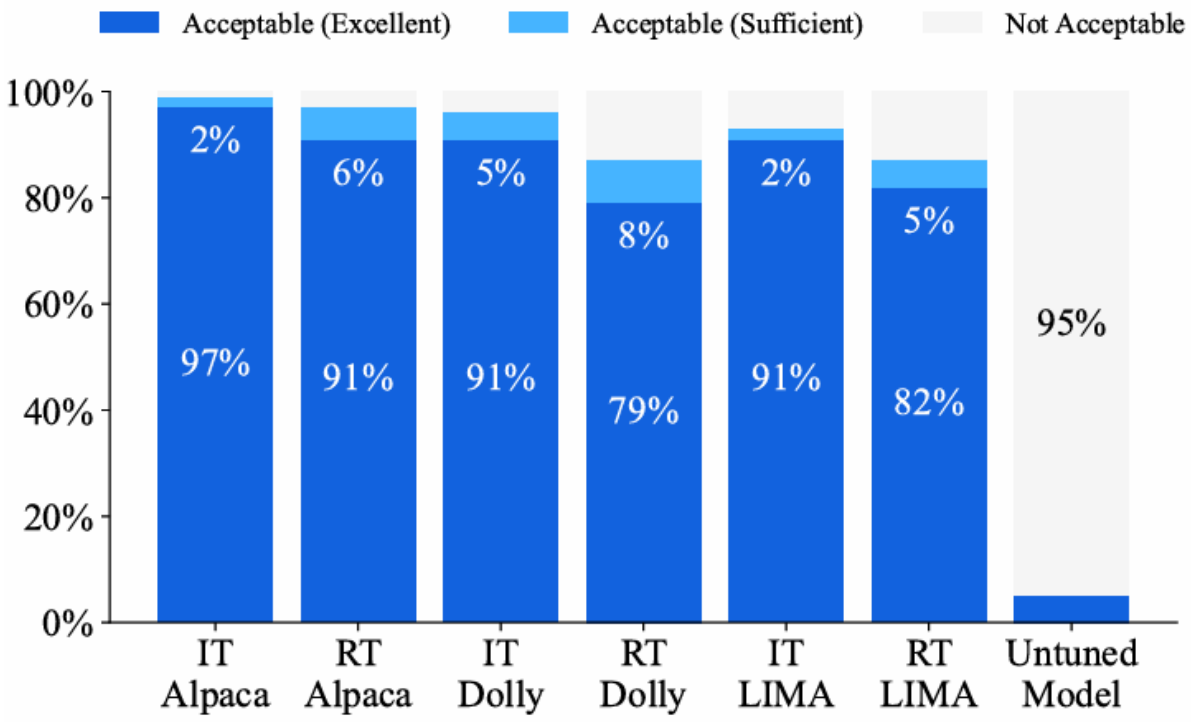
- **Response Tuning**

既然内容“不重要”，那就只调整回答的语气如何？

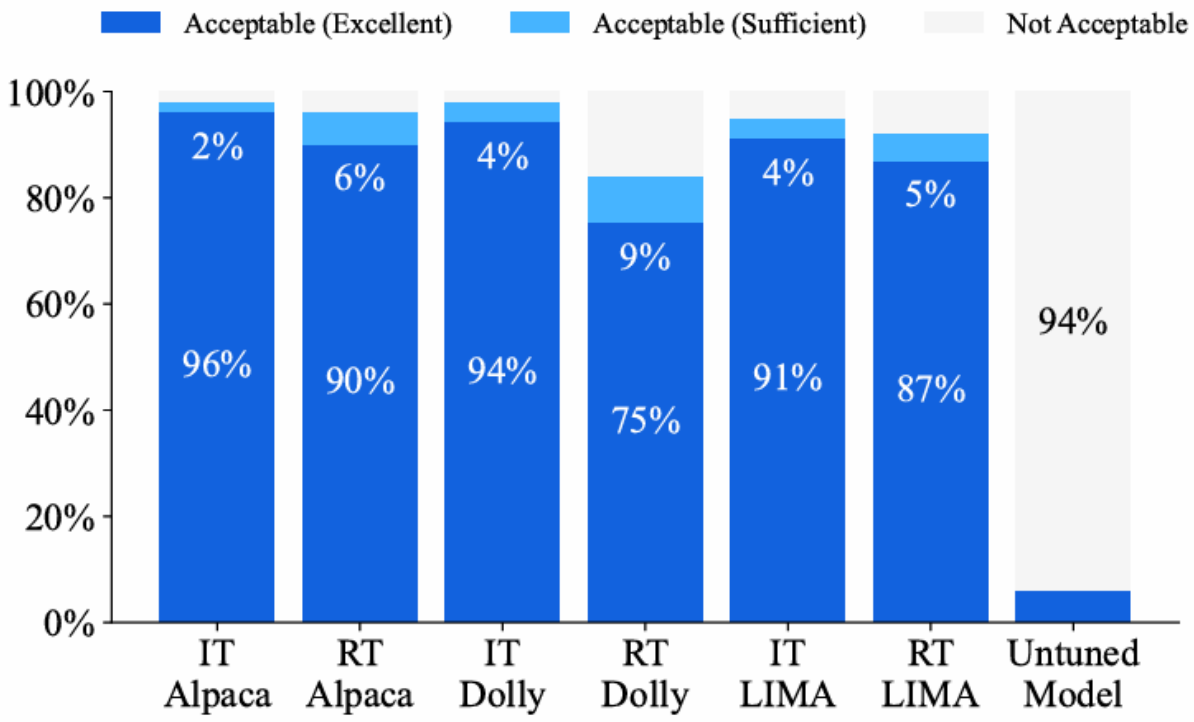
https://arxiv.org/abs/2410.02465v2

2.6 其实对齐的作用并没有那么大

- Response Tuning



(a) Base LLM: Llama-3.1-8B (Dubey et al., 2024)

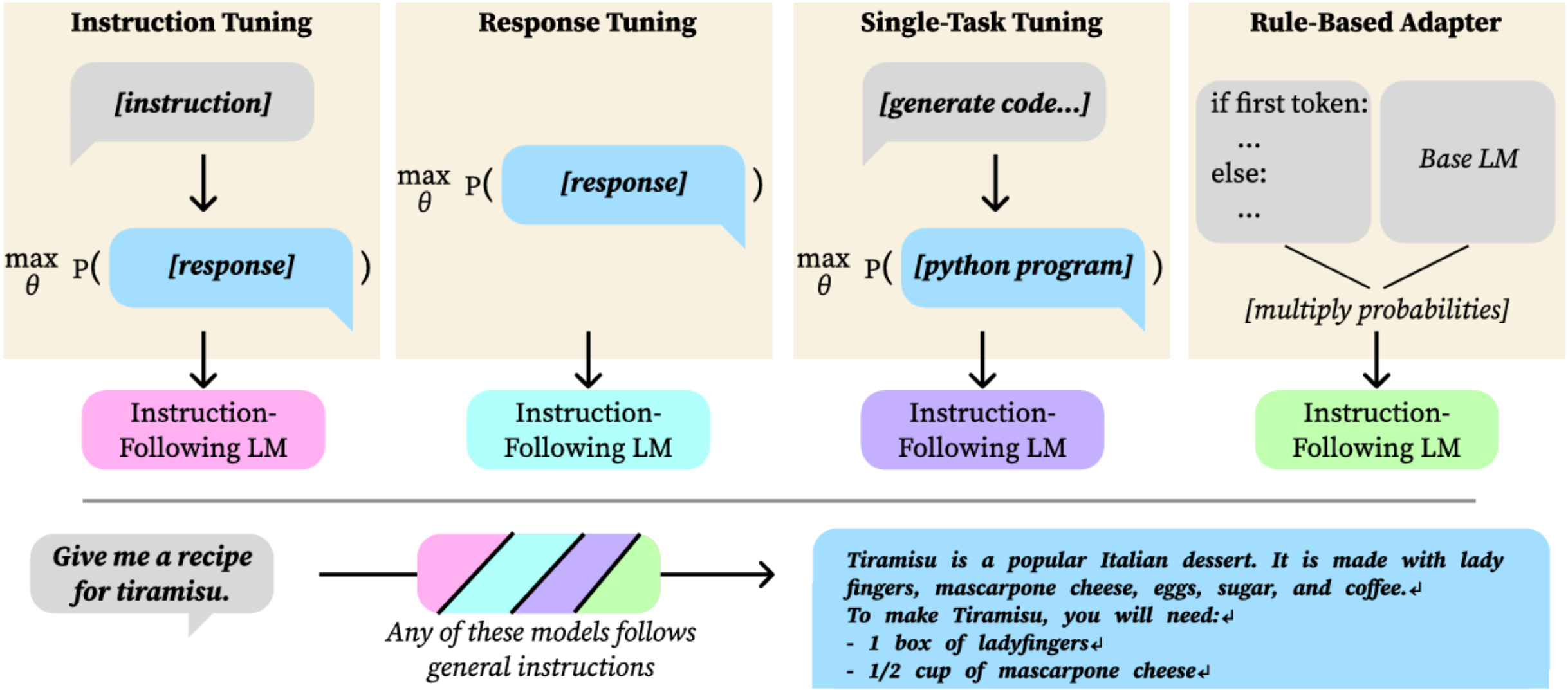


(b) Base LLM: Gemma-2-9B (Riviere et al., 2024)

https://arxiv.org/abs/2410.02465v2

2.6 其实对齐的作用并没有那么大

手工调整



https://arxiv.org/abs/2409.14254

2.6 其实对齐的作用并没有那么大

增加结束符号的概率

手动改变符号的概率

避免出现重复的符号

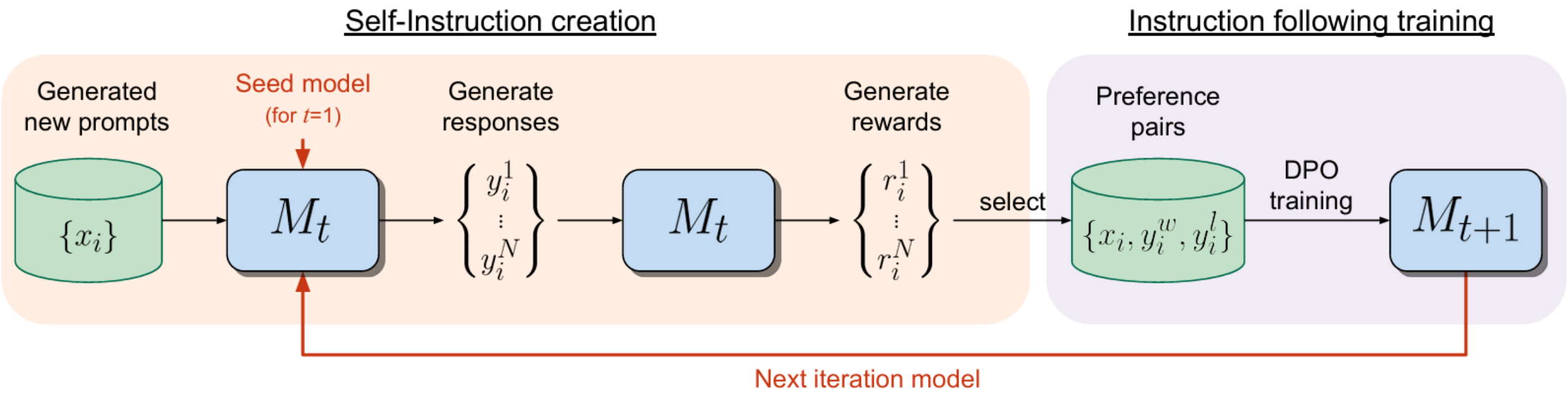
Rule	Vocab Items (string)	Weight
Rule 1 (Upweight EOS)	</S> (EOS)	$\frac{(\text{length of response})*15}{250}$
Rule 2 (Uniform Token Changes)	<, _<, |	-4
	_I, I	-5
	We	-3
	What	-3
	_should	-6
	_*, _-, ____, _#, _##, \n, !	+1
Rule 3 (Penalize Used Words)	$\{x \in \mathcal{V} \mid x \in \text{(response so far)}\}$	-1.5

Model	Rule-Based Model	Win Rate vs. Instruction Tuning
Llama-2-7B	None (Base)	$2.4\% \pm 0.14\%$
	All Rules	$24.4\% \pm 0.40\%$
	- EOS Rule (Rule 1)	$10.4\% \pm 0.30\%$
	- Diversity Rule (Rule 3)	$14.3\% \pm 0.58\%$
	- uniform token changes (Rule 2)	$16.3\% \pm 0.25\%$

https://arxiv.org/abs/2409.14254

2.6 其实对齐的作用并没有那么大

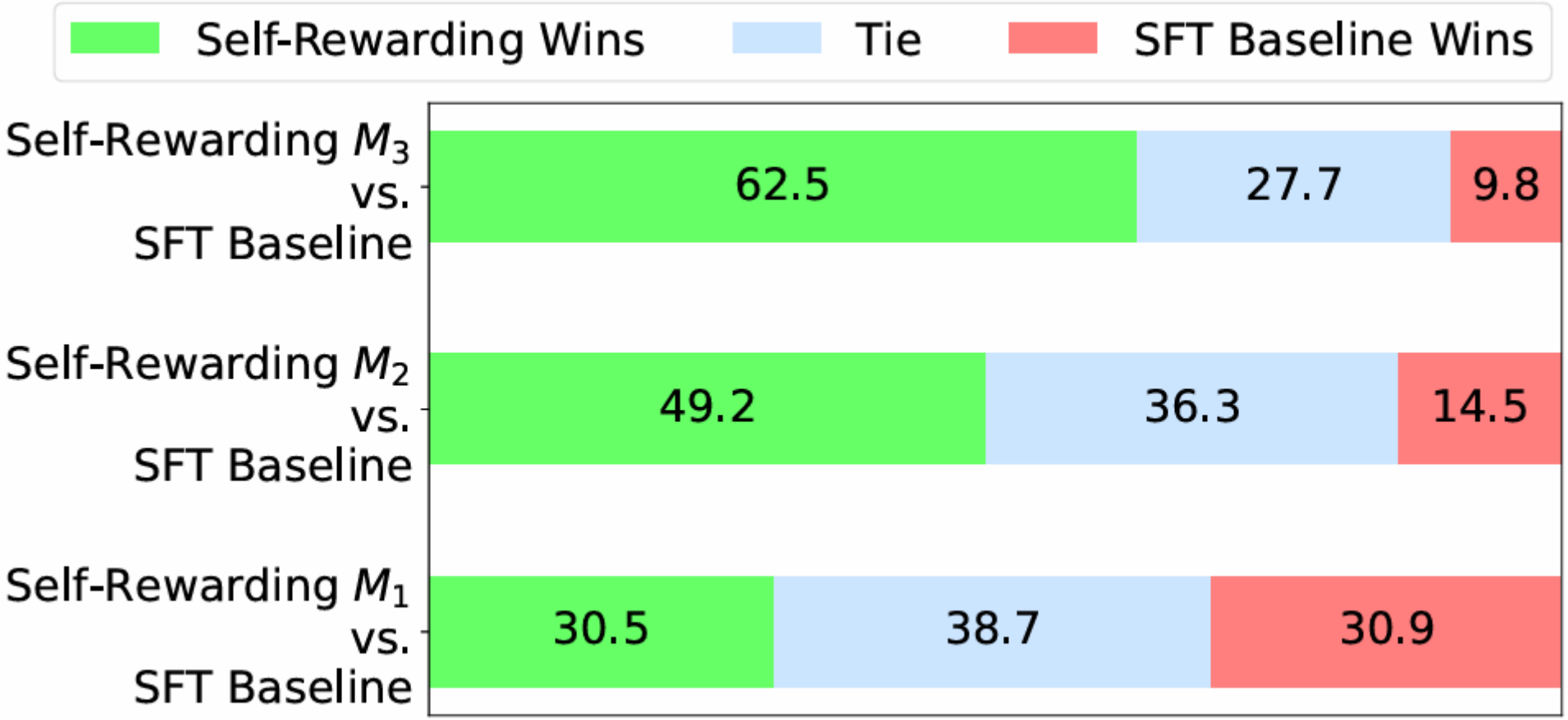
- **Self-alignment**



https://arxiv.org/abs/2401.10020

2.6 其实对齐的作用并没有那么大

- **Self-alignment**



https://arxiv.org/abs/2401.10020